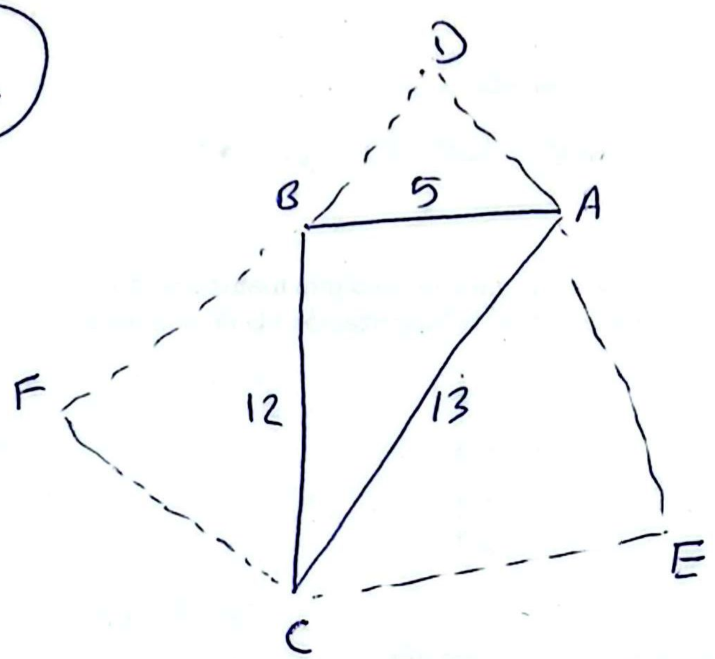
1

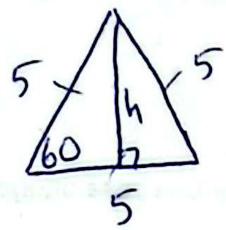


$AC^2 = AB^2 + BC^2$

$AC^2 = 5^2 + 12^2$

$\boxed{AC = 13}$

triangulo ADB



$\operatorname{Sen} 60 = \frac{h}{5} \Rightarrow h = 5 \operatorname{Sen} 60$

Luego $\text{Area}\,1 = \frac{1}{2} b \cdot h$

$= \frac{1}{2}(5)(5 \operatorname{Sen} 60)$

$= \frac{25}{2} \cdot \frac{\sqrt{3}}{2} = \frac{25}{4}\sqrt{3}$

Triangulo AEC: $\text{Area}\,2 = \frac{1}{2} b \cdot h$

$= \frac{1}{2}(13)(13 \operatorname{Sen} 60)$

$= \frac{169\sqrt{3}}{4}$

Triangulo BCF: $\text{Area}\,3 = \frac{1}{2} b \cdot h$

$= \frac{1}{2}(12)(12 \operatorname{Sen} 60)$

$= \frac{144}{4}\sqrt{3}$

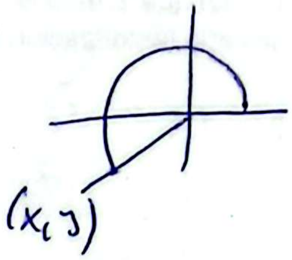
Por lo tanto

$\text{Area Total} = A1 + A2 + A3 = \frac{25\sqrt{3}}{4} + \frac{169\sqrt{3}}{4} + \frac{144\sqrt{3}}{4} = \boxed{\frac{169\sqrt{3}}{2}}$

Sobre el ángulo

$$\pi \le \alpha \le \frac{3}{2}\pi \Rightarrow \boxed{\frac{\pi}{2} \le \frac{\alpha}{2} \le \frac{3}{4}\pi}$$

$\therefore \quad \frac{\alpha}{2} \in$ II-Cuadrante



$$tg(\alpha) = \frac{12}{5} = \frac{y}{x} \Rightarrow \begin{cases} x=-5 \\ y=-12 \\ r=? \end{cases} \Rightarrow \begin{array}{l} r^2 = x^2+y^2 \\ r^2=(-5)^2+(-12)^2 \\ r=13 \end{array}$$

Luego $\quad tg\,\alpha = tg\left(2\cdot\frac{\alpha}{2}\right)$

$$\frac{12}{5} = \frac{2\cdot tg\left(\frac{\alpha}{2}\right)}{1-tg^2\left(\frac{\alpha}{2}\right)}$$

$$12\left(1-tg^2\left(\frac{\alpha}{2}\right)\right) = 10\,tg\left(\frac{\alpha}{2}\right)$$

$$6\,tg^2\left(\frac{\alpha}{2}\right) + 5\,tg\left(\frac{\alpha}{2}\right) - 6 = 0$$

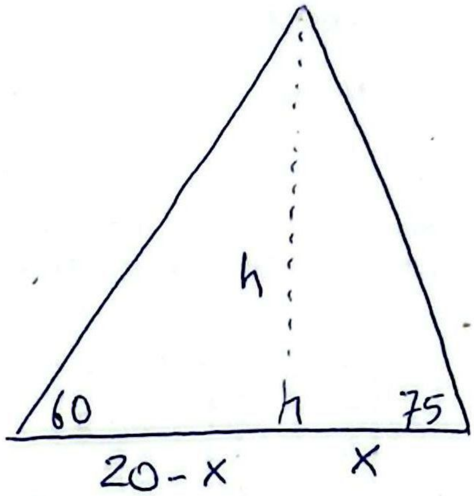
$$\left(3\,tg\frac{\alpha}{2} - 2\right)\cdot\left(2\,tg\frac{\alpha}{2}+3\right) = 0$$

Para $\ast)\ 3\,tg\frac{\alpha}{2} - 2 = 0 \Rightarrow tg\frac{\alpha}{2} = \frac{2}{3}$

$\ast)\ 2\,tg\frac{\alpha}{2} + 3 = 0 \Rightarrow tg\frac{\alpha}{2} = -\frac{3}{2}$

Como $\frac{\alpha}{2} \in II \Rightarrow \boxed{tg\frac{\alpha}{2} = -\frac{3}{2}}$

3



$$\tan 75 = \frac{h}{x} \qquad \Rightarrow \qquad \begin{cases} x = \dfrac{h}{tg\,75} \quad \cdots\cdots \quad ① \\ (20-x)\,tg\,60 = h \end{cases}$$

$$tg\,60 = \frac{h}{20-x}$$

$$20\,tg\,60 - x\,tg\,60 = h \quad \cdots ②$$

① en ②

$$20\,tg\,60 - \left(\frac{h}{tg\,75}\right) tg\,60 = h$$

$$20\,tg\,60 = \left(1 + \frac{tg\,60}{tg\,75}\right) h$$

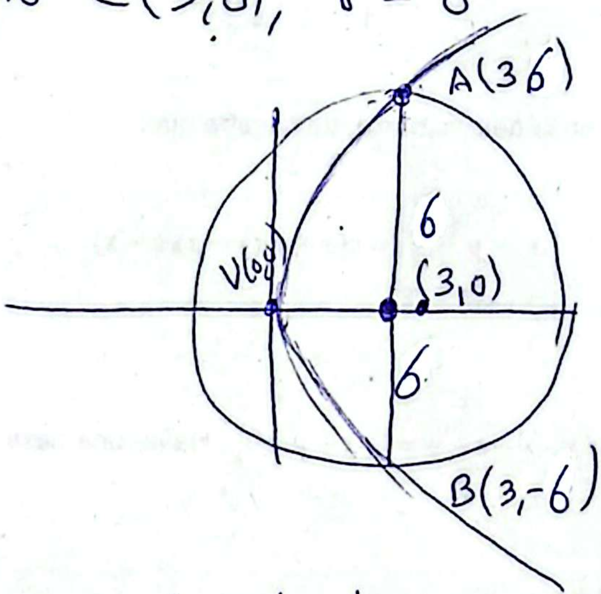
$$h = 15 + 5\sqrt{3}$$

$$\boxed{h = 23{,}66}$$

4

$$x^2+y^2-6x-27=0$$
$$x^2-6x+y^2=27$$
$$x^2-6x+3^2+y^2=27+3^2$$
$$(x-3)^2+(y-0)^2=6^2$$

Centro $C(3,0)$, $r=6$



La ecuación de la parábola de vertice el origen y eje X es: $y^2=4p\,x$

$A(3,6) \in$ Parábola $\Rightarrow 6^2=4p(3)$

$$p=3$$

Luego

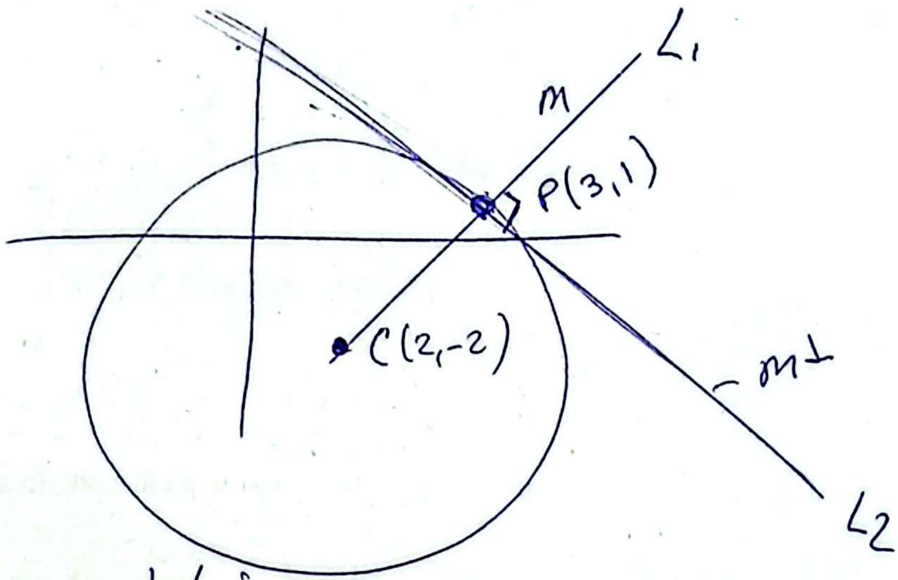
$$y^2=4(3)x$$

$$\boxed{y^2=12x}$$

(5)

$$x^2+y^2-4x+4y-2=0$$
$$x^2-4x \quad +y^2+4y \quad =2$$
$$x^2-4x+2^2+y^2+4y+2^2=2^2+2^2+2^2$$
$$(x-2)^2+(y+2)^2=12$$

Centro $C(2,-2)$, $r=\sqrt{12}$



Pendiente de L_1:

$$m=\frac{y_2-y_1}{x_2-x_1}=\frac{1-(-2)}{3-2} \Rightarrow \boxed{m=3}$$

Luego $m \cdot m^\perp=-1 \Rightarrow 3 \cdot m^\perp=-1 \Rightarrow \boxed{m^\perp=-\frac{1}{3}}$

La recta L_2 pasa por $P(3,1)$ con pendiente $m=-\frac{1}{3}$

es:

$$y-y_0=m(x-x_0)$$
$$y-1=-\frac{1}{3}(x-3)$$
$$3(y-1)=-(x-3)$$
$$\boxed{x+3y-6=0}$$